

Designing Interactive Experiences For Gut Health Engagement and Reflection

Nandini Pasumarthy

HAFP Research Lab, School of Design, RMIT University

Melbourne, VIC, Australia

s3826394@student.rmit.edu.au

ABSTRACT

Human gut health is the epicentre of human health and well-being. We engage in microscopic interactions in our day-to-day lives that influence our gut health, however, our understanding of this relationship is scarce. Current approaches to engage people on gut-related factors are heavily jargonised, lacking real-world application and focus on disease-causing aspects, thus limiting motivation to engage with gut health. Research suggests that games can act as powerful tools for engagement and reflection on this topic. This PhD research explores the design of two games to understand the key game design features that enable engagement with gut health. The results from testing these games will be generalised to inform the design of physical and digital games for engagement and reflection on health.

CCS CONCEPTS

• **Human-centered computing** → *Human computer interaction (HCI)*; Interactive systems and tools;

KEYWORDS

serious games, persuasive games, gut health, gut microbiome, human-food interaction

ACM Reference Format:

Nandini Pasumarthy. 2023. Designing Interactive Experiences For Gut Health Engagement and Reflection. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3544549.3577058>

1 BACKGROUND AND MOTIVATION

This PhD explores the design of interactive experiences to engage players in reflection on the factors that influence gut microbial diversity. Our gastrointestinal health is influenced by complex interactions between our gut bacteria and multiple external factors such as mode of birth, early microbial exposure, antibiotics, diet, lifestyle, physical activity, environment and even our emotions [57]. A wider understanding of these factors is vital to help make gut-friendly decisions in everyday life. The science behind gut functions is dynamic and constantly evolving in addition to the microscopic nature of the gut microbiome which makes gut health an even more

complex topic, challenging for people to understand and relate to especially if not approached in a systematic manner [2].

Research suggests that interactive experiences generated by play-based approaches can serve as powerful tools to deal with real-world problems and offer tremendous opportunities to engage people on various topics and in various contexts [7, 19, 28, 32, 34, 41, 53]. Games can be useful to engage people in health information by embedding real-world concepts into game experiences for real-world understanding [44]. Games and play are not only well recognised and used but are also considered as highly appropriate tools to trigger and support reflection on various topics [19, 23, 45, 60, 64, 72]. Designing games to deconstruct gut health information for public understanding, can particularly benefit from approaching it through the lens of reflection for engaging users in real world contextual understanding of the topic.

There has been a growing trend in using play to engage people in health science topics [8, 15, 26, 54]. The field of Human-Computer Interaction (HCI) and Human-Food Interaction (HFI) has seen various attempts at creating novel playful systems to communicate about complex topics within the field of science and technology through serious games, purposeful games, transformative and critical games for learning, engagement, reflection, and behaviour change among others [8, 54, 55, 71]. However, games on gut health are not only underexplored, but are also not designed to engage users in real-world understanding of the factors and their influence on gut health. Games related to the gut microbiome focus on achieving pedagogical outcomes for microbiology learning [1, 3, 16, 20], learning of scientific names and classes of bacteria with low or no emphasis on the real world influence of the factors on gut microbial diversity and gut health. There is also a lack of a playful hands-on approach for real world engagement on this topic. Overall, the topic of the gut-microbiome and gut health from a play perspective is underexplored, which is the focus of my research.

1.1 Research questions and objectives

My PhD explores the design of two game prototypes as part of two individual case-studies involving a) structured play through the design of a physical board game, and b) unstructured (i.e., open-ended) play through the design of a smartphone game. As a first step, I used structured (rule-based) play to engage users in gut health information through real world scenarios introduced by the board game mechanics. The findings from testing this approach, led to questions on engaging users in real-world contexts and physical activities through game play. To facilitate physical activity engagement in real world contexts, I explore unstructured (open-ended) play for the second case study, involving a smartphone game. The design and study of these two games will answer the primary and secondary research questions: “How do we design interactive experiences for

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '23, April 23–28, 2023, Hamburg, Germany

© 2023 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3577058>

engagement with, and reflection on gut health factors?” Sub research questions: Sub RQ1: *“How do we design structured play for engagement and reflection on gut health factors?”* Sub RQ2: *“How do we design open-ended play for engagement and reflection on gut health factors?”* These questions will address the following objectives:

- Identify the design space for interactive technologies and experiences for gut health engagement and reflection by mapping related literature and presenting two design prototypes.
- Validate the design prototypes through user study and its evaluation.
- Using the insights gathered, generate a design framework that aids designers in designing interactive experiences through structured and unstructured play for engagement, and reflection on gut health factors.

2 RELATED WORK

Prior work that directly addressed one or more gut health factors include health interfaces and applications to track specific digestive disorders such as irritable bowel syndrome (IBS) and leaky gut [4]. Some of these include wearable technologies to track bodily feedback [11, 69, 70], diet-based applications to journal and track food consumption patterns have been explored for engaging users on gut health Your Digital Platform to Better Gut Health. Play-based applications for encouraging healthy chewing practices [24], mindful-eating behaviours [9, 24, 38], and nutritional decision making [14, 49] use social mechanics through friendly social challenges and positive message framing to motivate nutritional behaviours. Serious games for health within and outside HCI have been explored to engage players on topics including human physiology and health promotion for topic understanding and behaviour change [10, 21, 28, 32, 66]. Playful visualisations and engaging people’s curiosity to peek into the human body and experience the inner mechanisms has been enabled through interactive gamified maps [1]. This includes serious health simulation games on gut bacteria [6] that offer immersive experiences on life inside the gut for playful visualisations and enable closer relatability to microscopic aspects of human physiology. Serious games for pedagogy and learning microbiology include both analogue [3, 16] and digital games [1], DIY citizen science games that invite players to engage and solve real world science challenges through play to help researchers include sorting through digital data of stool samples to assist scientists in the identification and clustering of microbial colonies [34, 67]. These games involve users in play and enable the creation of meaningful play experiences by exposing users to abstract real world data for problem-solving.

Embodied approaches to experience more-than-human interactions have been carried out through speculative and scientific experiments using agar plate inoculation for engaging and tracking human-microbial interactions using live microbial cultures and fermentation experiments [13, 22]. These material explorations use tangible natural materials to understand human-microbial interaction thus offering an embodied experience of tinkering with natural processes. Furthermore, they also leverage on the uncertainty involved in working with microbes where the outcomes may not always be predictable, and thus embrace the ambiguity that

such natural processes bring with it. Such experiments with live cultures have also led researchers within HCI to explore the design of biotic games involving the integration of microbes with computation thus offering players a dive into the microbial realm that contain beneficial, opportunistic and pathogenic bacteria that make interesting and playful narratives [5, 39, 40]. These embodied experiences can be useful to spur curiosity and reflection on the presence and role of microbes in our everyday life, however they do not engage users on the benefits of the factors and their influence on gut microbial diversity. Inspired by the slowness involved in bodily process, and previous works that emphasise on slowness for reflection, I explore time-based mechanics and investigate the design of slow technologies for reflection. Exploratory technologies for domestic use that employ slowness as a design feature have been successful in encouraging reflection on everyday life [47], digestive processes [37], nature [7], and food interactions [33]. In relation to reflection, understanding the role of technology as a mediator to learn about the subtler aspects of life such as our digestive processes [37], gut microbiome [51], and plant life [59] as influenced by post-phenomenological approach inspires my research [68].

Interactive experiences generated by games and play hold an important role in enabling reflection. However, as scholars suggest, reflection cannot merely be an outcome of the design, but that which is deliberately intertwined with it, and in which both the designer and the user are actively involved in [27, 62]. Existing games and interactive applications on gut health are not designed to engage players in reflection but rather focus on achieving pedagogical outcomes for microbiology learning. I learn from the pool of related work on gut health and microbiome [3, 16], digital games [1], embodied interaction [13, 22], science engagement [5, 39, 40], and mechanics and strategies for reflection [27, 36, 62] to design interactive experiences using structured and unstructured play for gut health reflection.

3 METHODS

3.1 Research through Design (RtD)

I utilise research through design (RtD) which is a widely used method in technology, art, and engineering-based design research [43] (p.39). RtD is a reflective and iterative approach to design involving a cycling process of designing, prototyping, and testing until a desired level of outcome is achieved. RtD involves an exploration into why and how people interact with an artefact and the context in which these interactions occur [25, 42]. According to Koskinen, research contributions should not be limited to artefact production alone. Instead, it must be accompanied by a critical reflection on the process, the design decisions and the designer’s reflection on those decisions thus offering future designers an insight into the design specifications and guidelines to design similar artefacts within the specific context [29, 30, 42]. This is in line with the argument made by Sengers and Khaled on the designer’s role on reflecting on the design while designing for reflection [36, 62]. Game design as such is an iterative design process and the use of research through design (RtD) methodology is deemed appropriate within the field of HCI.

3.2 Mechanics, Dynamics, and Aesthetics Framework (MDA)

I used Mechanics, Dynamics, and Aesthetics (MDA) [35], a popular game design framework to design and define the game features for both the games. Initially, I started by mapping the subject matter, concepts and information that was important for players to understand and take back into the real-world post play. Next, I identified the key aesthetics that I wanted players to experience while interacting with the game including feelings of surprise, curiosity, stewardship, empathy, frustration, excitement, and relatedness to enable an emotional connection to gut microbes and factors. I used the deck of lenses by Jesse Schell to develop this [61]. Following that, I identified a set of interactions and intertwined the game content to a set of game mechanics and dynamics that lead to corresponding aesthetics.

3.3 Integrated Model for Open-ended Play (IMO)

I used the Integrated Model for Open-Ended Play (IMO) [56] to design the second case-study that involves a digital smartphone game for unstructured play. The IMO is an extension of the MDA framework and Grünvogel's Formal Model of Game design [31]. This framework involves viewing the game design from two perspectives namely the perspective of design, that involves the designed mechanics and corresponding dynamics as intended by the game designer which is described using the terms defined progression and inner emergence. Secondly, the perspective of play, which depicts the need for creating a space for users to construct their own rules in an open-ended game, described through the terms- user-defined progression, and outer emergence where play occurs outside of the rules generated by the designer, and based on the new rules set by the player. Doing so, the framework emphasises on the need to make space for open-ended play by way of inner and outer emergence, meaning, play that occurs within and outside the mechanics set by the designer. I use this framework to explore the design of open-ended play for adults.

I used in-the-wild study [58] and qualitative research approach involving pre-study and post-study interviews in semi-structured format to gather user awareness on gut health factors, their experiences with the game features, and corresponding feedback. Additionally, observations of video data were conducted from the board game playtest study sessions for the first case study. The second case study includes interviews, a questionnaire and photo stories from the participants. Photo stories as a cultural probe for documentation have previously been used in empirical studies to gather context-related information, especially in the absence of sensors to gather location and context specific details in digital games [48, 50, 65].

3.4 Understanding player experience through the lens of reflection

Data gathered from the interviews will be coded thematically and analysed through the lens of reflection [27, 36, 62]. Reflection as a core design element is increasingly popular in HCI for designing interactive technologies [12, 17, 23, 33, 37, 47, 63]. Research suggests

that reflection plays a crucial role in making people self-aware of their actions that influence their life and those around them. Reflection involves revisiting and re-assessing one's own beliefs and experiences intentionally and proactively. However, reflection may not come naturally to everyone, and it is a process that often needs to be facilitated [36]. Previous work on reflection within HCI suggests that reflection cannot be just an outcome of design, but rather it needs to be thoughtfully intertwined into the design of the artefact or experience [46]. This led to the development of design strategies for reflection and design qualities of reflective design [36, 62]. Based on these theories of reflection and reflective design, I approach the design, its documentation, and the user-experience analysis of the two case studies through the lens of reflection [18, 36, 62]. In doing so, my research focuses on designing interactive experiences through play to support reflection on gut health factors. The findings from the two case studies will inform a design framework including a set of design strategies for designing interactive experiences for engagement and reflection on gut health using structured and unstructured play approaches.

4 DISSERTATION STATUS AND NEXT STEPS

The initial investigation commenced through the design of a board game called Goody Gut Trail (Figure 1) that engaged users in gut health understanding. The game was tested through a field study with 15 participants. The user study evaluation resulted in 12 findings which detail different game mechanics and their corresponding user-experiences. The findings led to the articulation of 6 design strategies to design games for gut health engagement and reflection. The design documentation has been published through a pictorial at Creativity and Cognition 2021 [52]. A full paper detailing the study and findings has been published at CHI Play 2022 [51]. Based on the feedback received from the first case study, the second case study explores the design of Go-Go Biome (Figure 2), a smartphone game that involves users in real-world physical activity engagement for reflection on gut health factors. The game introduces physical, environmental, social and mindfulness activities in the form of activity card decks for users to explore and engage in. By engaging in diverse activities, players witness the growth of friendly biome and the decrease of unfriendly biome population as they attempt to balance the two to a ratio of 85:15 (friendly: unfriendly) biome population. The field study is currently in progress, and this may inspire the design of a third case study (not yet designed), and thus inform the final design framework. Overall, I see this research as a critical step towards understanding the design of interactive experiences for health engagement and reflection which can have greater implications for future health applications.

5 EXPECTED CONTRIBUTIONS

My research will make the following contributions: 1) artefact contributions through the design and development of games as interactive technologies for engagement and reflection on gut health factors, 2) theoretical contributions that extend our understanding of designing unstructured play that motivate player reflection on gut health factors, and 3) empirical contributions through the two user studies that reveal player experiences. The associated findings



Figure 1: Case Study 1- Gooney Gut Trail, a board game designed to engage players on gut health factors.

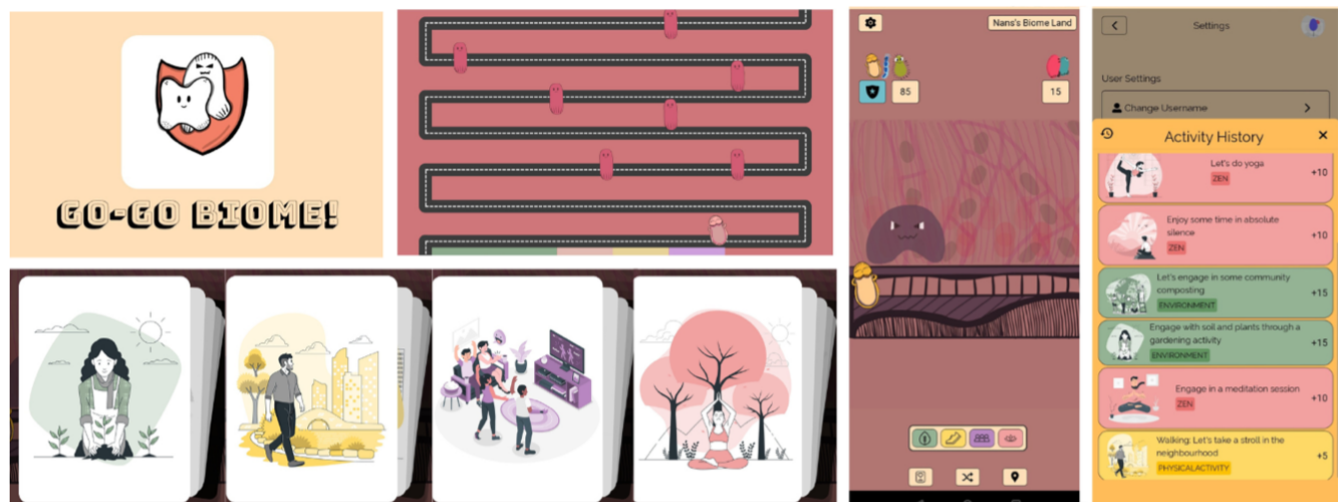


Figure 2: Case Study 2- Go-Go Biome, a smartphone game to engage players in physical activities for gut health.

will be distilled into themes that can aid future playful explorations on engaging with gut health.

REFERENCES

- [1] 2016. <https://www.amnh.org/explore/ology/microbiology/human-microbiome-minecraft-map>
- [2] 2017. <https://www.newhope.com/health-and-nutrition-research/consumers-interested-gut-microbiome-lack-understanding>
- [3] 2017. <https://www.amnh.org/explore/ology/microbiology/gutsty-the-gut-microbiome-card-game>
- [4] 2019. <https://bowelle.com/>
- [5] 2020. <https://interactions.acm.org/archive/view/march-april-2020/toward-growable-computer-games>

- [6] 2021. https://store.steampowered.com/app/1676620/Kelloggs_Gut_Bacteria_Reef/
- [7] Riga Anggarendra and Margot Brereton. 2016. Engaging children with nature through environmental HCI. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction (OzCHI '16)*. Association for Computing Machinery, New York, NY, USA, 310–315. <https://doi.org/10.1145/3010915.3010981>
- [8] Emeka G. Anyanwu. 2014. Anatomy adventure: a board game for enhancing understanding of anatomy. *Anatomical sciences education* 7, 2 (Mar 2014), 153–160. <https://doi.org/10.1002/ase.1389>
- [9] Eshita Sri Arza, Harshitha Kurra, Rohit Ashok Khot, and Florian “floyd” Mueller. 2018. Feed the Food Monsters! Helping Co-diners Chew their Food Better with Augmented Reality. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts (CHI PLAY '18 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 391–397. <https://doi.org/10.1145/3270316.3271520>
- [10] Lance Baily and HealthySimulation.com Writing Team. 2021. Top Healthcare Simulation Serious Games to Engage Clinical Learners. <https://www.healthysimulation.com/35309/top-serious-games/>
- [11] Annalisa Baronetto, Luisa S. Graf, Sarah Fischer, Markus F. Neurath, and Oliver Amft. 2020. GastroDigitalShirt: a smart shirt for digestion acoustics monitoring. In *Proceedings of the 2020 International Symposium on Wearable Computers (ISWC '20)*. Association for Computing Machinery, New York, NY, USA, 17–21. <https://doi.org/10.1145/3410531.3414297>
- [12] Marit Bentvelzen, Paweł W. Woźniak, Pia S. F. Herbes, Evropi Stefanidi, and Jasmin Niess. 2022. Revisiting Reflection in HCI: Four Design Resources for Technologies that Support Reflection. *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 6, 1 (Mar 2022), 1–27. <https://doi.org/10.1145/3517233>
- [13] Laurens Boer, Harvey Bewley, Tom Jenkins, Sarah Homewood, Teresa Almeida, and Anna Vallgård. 2020. Gut-Tracking as Cultivation. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 561–574. <https://doi.org/10.1145/3357236.3395588>
- [14] Jennifer Boothroyd. 2017. *What's on My Plate?: Choosing from the Five Food Groups*. Lerner Digital TM. <https://play.google.com/store/books/details?id=HtsiEAAAQBAJ>
- [15] Julio Cesar Queiroz de Cavalho, Leila Maria Beltramini, and Nelma Regina Segnini Bossolan. 2019. Using a board game to teach protein synthesis to high school students. *Journal of biological education* 53, 2 (Mar 2019), 205–216. <https://doi.org/10.1080/00219266.2018.1469532>
- [16] David A. Coil, Cassandra L. Ettinger, and Jonathan A. Eisen. 2017. Gut Check: The evolution of an educational board game. *PLoS biology* 15, 4 (Apr 2017), e2001984. <https://doi.org/10.1371/journal.pbio.2001984>
- [17] Alma Leora Culén and Katie Coughlin. 2022. Growing Up in a Complex World: Engaging Children in Socio-Cultural Matters Through Speculative Installations. In *Designing Interactive Systems Conference (DIS '22)*. Association for Computing Machinery, New York, NY, USA, 693–706. <https://doi.org/10.1145/3532106.3533518>
- [18] Peter Dalsgaard and Kim Halskov. 2012. Reflective design documentation. In *Proceedings of the Designing Interactive Systems Conference (DIS '12)*. Association for Computing Machinery, New York, NY, USA, 428–437. <https://doi.org/10.1145/2317956.2318020>
- [19] Joshua A. Danish, Noel Enyedy, Asmalina Saleh, Christine Lee, and Alejandro Andrade. 2015. Science through Technology Enhanced Play: Designing to support reflection through play and embodiment. (Jul 2015). <https://repository.isls.org/handle/1/425>
- [20] Rafael Paschoal de Campos, Victória Pereira Viero, Natascha Monteiro Medeiros, Fernanda Klein Marcondes, Luis Henrique Montezor, Marilene Porawski, and Lucila Ludmila Paula Gutierrez. 2020. The “Gut Game”: an active methodology to teach digestive physiology. *Advances in physiology education* 44, 3 (Sep 2020), 444–447. <https://doi.org/10.1152/advan.00007.2020>
- [21] Jéssica David Dias, Marcelo Shinyu Mekaro, Jennifer Kaon Cheng Lu, Joice Lee Otsuka, Luciana Mara Monti Fonseca, and Silvia Helena Zem-Mascarenhas. 2016. Serious game development as a strategy for health promotion and tackling childhood obesity. *Revista latino-americana de enfermagem* 24 (Aug 2016), e2759. <https://doi.org/10.1590/1518-8345.1015.2759>
- [22] Markéta Dolejšová, Danielle Wilde, Ferran Altarriba Bertran, and Hilary Davis. 2020. Disrupting (More-than-) Human-Food Interaction: Experimental Design, Tangibles and Food-Tech Futures. In *Proceedings of the 2020 ACM Designing Interactive Systems Conference (DIS '20)*. Association for Computing Machinery, New York, NY, USA, 993–1004. <https://doi.org/10.1145/3357236.3395437>
- [23] Tao Dong, Mark W. Newman, Mark S. Ackerman, and Sarita Schoenebeck. 2015. Supporting reflection through play: field testing the home trivia system. In *Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*. Association for Computing Machinery, New York, NY, USA, 251–262. <https://doi.org/10.1145/2750858.2804294>
- [24] Daniel A. Epstein, Felicia Cordeiro, James Fogarty, Gary Hsieh, and Sean A. Munson. 2016. Crumbs: Lightweight Daily Food Challenges to Promote Engagement and Mindfulness. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16, Vol. 2016)*. Association for Computing Machinery, New York, NY, USA, 5632–5644. <https://doi.org/10.1145/2858036.2858044>
- [25] Daniel Fallman. 2003. Design-oriented human-computer interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '03)*. Association for Computing Machinery, New York, NY, USA, 225–232. <https://doi.org/10.1145/642611.642652>
- [26] David Farrell, Patty Kostkova, Julius Weinberg, Lisa Lazareck, Dasun Weerasinghe, Donna M. Lecky, and Cliona A. M. McNulty. 2011. Computer games to teach hygiene: an evaluation of the e-Bug junior game. *The Journal of antimicrobial chemotherapy* 66 Suppl 5 (Jun 2011), v39–44. <https://doi.org/10.1093/jac/dkr122>
- [27] Rowanne Fleck and Geraldine Fitzpatrick. 2010. Reflecting on reflection: framing a design landscape. In *Proceedings of the 22nd Conference of the Computer-Human Interaction Special Interest Group of Australia on Computer-Human Interaction (OZCHI '10)*. Association for Computing Machinery, New York, NY, USA, 216–223. <https://doi.org/10.1145/1952222.1952269>
- [28] Jose M. Font, Alexander Hedvall, and Emil Svensson. 2017. Towards Teaching Maternal Healthcare and Nutrition in Rural Ethiopia through a Serious Game. In *Extended Abstracts Publication of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '17 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 187–193. <https://doi.org/10.1145/3130859.3131291>
- [29] Bill Gaver and John Bowers. 2012. Annotated portfolios. *Interactions* 19, 4 (Jul 2012), 40–49. <https://doi.org/10.1145/2212877.2212889>
- [30] William Gaver. 2012. What should we expect from research through design?. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '12)*. Association for Computing Machinery, New York, NY, USA, 937–946. <https://doi.org/10.1145/2207676.2208538>
- [31] S. Grünvogel. 2005. Formal Models and Game Design. *Game Stud.* (2005). <https://www.semanticscholar.org/paper/9ebfc1a30f41ab5d0601b0cd2470dceff0f1258f4>
- [32] Victor Guana, Tracy Xiang, Hannah Zhang, Ella Schepens, and Eleni Stroulia. 2014. UnderControl an educational serious-game for reproductive health. In *Proceedings of the first ACM SIGCHI annual symposium on Computer-human interaction in play (CHI PLAY '14)*. Association for Computing Machinery, New York, NY, USA, 339–342. <https://doi.org/10.1145/2658537.2662983>
- [33] Lars Hallnäs and Johan Redström. 2001. Slow Technology – Designing for Reflection. *Personal and Ubiquitous Computing* 5, 3 (Jan 2001), 201–212. <https://doi.org/10.1007/PL00000019>
- [34] Taylor Hatmaker. 2020. Borderlands 3 bridges the gap between citizen science and blockbuster games. *TechCrunch* (Apr 2020). <http://techcrunch.com/2020/04/07/borderlands-3-science-research-gut-microbiome/>
- [35] Robin Hunnicke, Marc LeBlanc, and Robert Zubek. 2004. MDA: A formal approach to game design and game research. In *Proceedings of the AAAI Workshop on Challenges in Game AI, Vol. 4*. 1722. <https://www.aaai.org/Papers/Workshops/2004/WS-04-04/WS04-04-001.pdf>
- [36] Rilla Khaled. 2018. *Questions Over Answers: Reflective Game Design*. Springer Singapore, Singapore, 3–27. https://doi.org/10.1007/978-981-10-1891-6_1
- [37] Rohit Ashok Khot, Jason Ng, and Deepti Aggarwal. 2022. Crafting Tangible Interfaces for Human Digestion: Unpacking the Research through Design Prototyping Journey. In *Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '22, Article 28)*. Association for Computing Machinery, New York, NY, USA, 1–15. <https://doi.org/10.1145/3490149.3502252>
- [38] Rohit Ashok Khot, Jung-Ying (lois) Yi, and Deepti Aggarwal. 2020. SWAN: Designing a Companion Spoon for Mindful Eating. In *Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '20)*. Association for Computing Machinery, New York, NY, USA, 743–756. <https://doi.org/10.1145/3374920.3375009>
- [39] Honesty Kim, Lukas C. Gerber, and Ingmar H. Riedel-Kruse. 2016. Interactive Biotechnology: Building your own Biotic Game Setup to Play with Living Microorganisms. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 1000–1002. <https://doi.org/10.1145/2851581.2856692>
- [40] Raphael Kim, Siobhan Thomas, Roland van Dierendonck, Nick Bryan-Kinns, and Stefan Poslad. 2020. Working with Nature's Lag: Initial Design Lessons for Slow Biotic Games. In *International Conference on the Foundations of Digital Games (FDG '20, Article 29)*. Association for Computing Machinery, New York, NY, USA, 1–4. <https://doi.org/10.1145/3402942.3409790>
- [41] Johanna Kleinen. 2018. Using Games and Play to Foster Collaboration and Innovation in a Cross-Organizational Working Environment. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts (CHI PLAY '18 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 27–31. <https://doi.org/10.1145/3270316.3270602>
- [42] Ilpo Koskinen, John Zimmerman, Thomas Binder, Johan Redström, and Stephan Wensveen. 2013. Design Research Through Practice: From the Lab, Field, and Showroom. *IEEE Transactions on Professional Communication* 56, 3 (2013), 262–263. <https://doi.org/10.1109/tpc.2013.2274109>
- [43] Peter Gall Krogh, Thomas Markussen, and Anne Louise Bang. 2015. Ways of Drifting—Five Methods of Experimentation in Research Through Design. In *ICORD'15 – Research into Design Across Boundaries Volume 1*. Springer India, 39–50. https://doi.org/10.1007/978-81-322-2232-3_4

- [44] Bernhard Maurer. 2017. Embodied Interaction in Play: Social and Physical Qualities as a Design Material for Digital Play. In *Extended Abstracts Publication of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '17 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 711–714. <https://doi.org/10.1145/3130859.3133223>
- [45] Elisa D. Mekler, Ioanna Iacovides, and Julia Ayumi Bopp. 2018. “A Game that Makes You Question...”: Exploring the Role of Reflection for the Player Experience. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '18)*. Association for Computing Machinery, New York, NY, USA, 315–327. <https://doi.org/10.1145/3242671.3242691>
- [46] Ine Mols, Elise van den Hoven, and Berry Eggen. 2016. Informing Design for Reflection: an Overview of Current Everyday Practices. In *Proceedings of the 9th Nordic Conference on Human-Computer Interaction (NordiCHI '16, Article 21)*. Association for Computing Machinery, New York, NY, USA, 1–10. <https://doi.org/10.1145/2971485.2971494>
- [47] Ine Mols, Elise van den Hoven, and Berry Eggen. 2017. Balance, Cogito and Dott: Exploring Media Modalities for Everyday-life Reflection. In *Proceedings of the Eleventh International Conference on Tangible, Embedded, and Embodied Interaction (TEI '17)*. Association for Computing Machinery, New York, NY, USA, 427–433. <https://doi.org/10.1145/3024969.3025069>
- [48] Elena Márquez Segura, Annika Waern, Jin Moen, and Carolina Johansson. 2013. The design space of body games: technological, physical, and social design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '13)*. Association for Computing Machinery, New York, NY, USA, 3365–3374. <https://doi.org/10.1145/2470654.2466461>
- [49] Samantha Olinsky, Pooja M. Desai, Selen Turkay, Elizabeth M. Heitkemper, Elliot G. Mitchell, Lena Mamykina, and Maria L. Hwang. 2021. *Meals for Monsters: a Mobile Application for the Feasibility of Gaming and Social Mechanisms*. Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3411763.3451789>
- [50] Katie O’Leary, Leslie Liu, Jennifer B. McClure, James Ralston, and Wanda Pratt. 2016. Persuasive Reminders for Health Self-Management. *AMIA ... Annual Symposium proceedings / AMIA Symposium* 2016 (2016), 994–1003. <https://www.ncbi.nlm.nih.gov/pubmed/28269896>
- [51] Nandini Pasumarthi, Rakesh Patibanda, Yi Ling (ellie) Tai, Elise van den Hoven, Jessica Danaher, and Rohit Ashok Khot. 2022. Goey Gut Trail: Board Game Play to Understand Human-Microbial Interactions. *Proc. ACM Hum.-Comput. Interact.* 6, CHI PLAY (Oct 2022), 1–31. <https://doi.org/10.1145/3549502>
- [52] Nandini Pasumarthi, Yi Ling (ellie) Tai, Rohit Ashok Khot, and Jessica Danaher. 2021. Goey Gut Trail :Demystifying Human Gut Health Through a Board Game. In *Creativity and Cognition (C&C '21, Article 19)*. Association for Computing Machinery, New York, NY, USA, 1–16. <https://doi.org/10.1145/3450741.3465390>
- [53] Daniel Pimentel. 2018. States of Mind, Body, and Planet: Improving the Human Condition through Virtual Experiences. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts (CHI PLAY '18 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 57–62. <https://doi.org/10.1145/3270316.3270609>
- [54] Kim Raphael, Dierendonck Roland van, and Poslad Stefan. 2019. Moldy Ghosts and Yeasty Invasions: Glitches in Hybrid Bio-Digital Games. (2019). <https://semanticscholar.org/paper/b08e4ad047fd1483974174a8cb26a9f54280677a>
- [55] Kim Raphael and Poslad Stefan. 2019. Growable, Invisible, Connected Toys: Twitching Towards Ubiquitous Bacterial Computing. (2019). <https://semanticscholar.org/paper/ee23dab03ffcf1b699cce5124e9887bb3e4321ba>
- [56] Pepijn Rijnbout, Mark de Graaf, Tilde Bekker, and Ben Schouten. 2016. The introduction of IMO, an integrated model for designing for open-ended play. *Entertainment computing* 16 (Jul 2016), 29–39. <https://doi.org/10.1016/j.entcom.2016.06.001>
- [57] Emanuele Rinninella, Pauline Raoul, Marco Cintoni, Francesco Franceschi, Giacinto Abele Donato Miggiano, Antonio Gasbarrini, and Maria Cristina Mele. 2019. What is the Healthy Gut Microbiota Composition? A Changing Ecosystem across Age, Environment, Diet, and Diseases. *Microorganisms* 7, 1 (Jan 2019). <https://doi.org/10.3390/microorganisms7010014>
- [58] Yvonne Rogers and Paul Marshall. 2017. Research in the Wild. *Synthesis Lectures on Human-Centered Informatics* 10, 3 (Apr 2017), i–97. <https://doi.org/10.2200/S00764ED1V01Y201703HCI037>
- [59] Margrete Lodahl Rolighed, Ester Marie Aagaard, Marcus Due Jensen, Raune Frankjaer, and Lone Koefoed Hansen. 2022. Plant Radio: Tuning in to plants by combining posthumanism and design. In *Designing Interactive Systems Conference (DIS '22)*. Association for Computing Machinery, New York, NY, USA, 666–676. <https://doi.org/10.1145/3532106.3533517>
- [60] Vanessa Santos and Roc Parés Burguès. 2018. Authoring a Serious Pervasive Game for Reflecting upon Urban Spaces. In *Interactivity, Game Creation, Design, Learning, and Innovation*. Springer International Publishing, 74–84. https://doi.org/10.1007/978-3-319-76908-0_8
- [61] Jesse Schell. 2008. *The Art of Game Design: A book of lenses*. CRC Press. <https://play.google.com/store/books/details?id=ZHFMBQAAQBAJ>
- [62] Phoebe Sengers, Kirsten Boehner, Shay David, and Joseph “jofish” Kaye. 2005. Reflective design. In *Proceedings of the 4th decennial conference on Critical computing: between sense and sensibility (CC '05)*. Association for Computing Machinery, New York, NY, USA, 49–58. <https://doi.org/10.1145/1094562.1094569>
- [63] Anjum Shaheen, Panagiotis Fotaris, and Sanaz Fallahkhair. 2021. A Systematic Review of Using Reflective Design Features in Game-Based Learning. In *15th European Conference on Game Based Learning ECGBL 2021*. unknown. <https://doi.org/10.34190/GBL.21.099>
- [64] Amanda C. R. Tavares, Rosilane Ribeiro da Mota, and Walter Melo. 2021. Self-reflection in Games - The Representation of the Individuation Process in Celeste and Persona 2: Innocent Sin. In *Anais Estendidos do XX Simpósio Brasileiro de Jogos e Entretenimento Digital*. SBC, 198–207. https://doi.org/10.5753/sbgames_estendido.2021.19640
- [65] Marije ten Brink, Tamara Witschge, Bert Bredeweg, and Ben Schouten. 2022. Designing for self-awareness: supporting students’ reflexive interactions based on photos. In *Creativity and Cognition (C&C '22)*. Association for Computing Machinery, New York, NY, USA, 161–170. <https://doi.org/10.1145/3527927.3532807>
- [66] Rosy Tsopra, Mélanie Courtine, Karima Sedki, David Eap, Manon Cabal, Samuel Cohen, Olivier Bouchaud, Frédéric Mechai, and Jean-Baptiste Lamy. 2020. AntibioGame®: A serious game for teaching medical students about antibiotic use. *International journal of medical informatics* 136 (Apr 2020), 104074. <https://doi.org/10.1016/j.ijmedinf.2020.104074>
- [67] McGill University. 2016. Gaming for gut research: New game advances research into microbes that play a role in our health. *Science Daily* (Aug 2016). <https://www.sciencedaily.com/releases/2016/08/160831162953.htm>
- [68] Peter-Paul Verbeek. 2015. COVER STORY Beyond interaction: a short introduction to mediation theory. *Interactions* 22, 3 (Apr 2015), 26–31. <https://doi.org/10.1145/2751314>
- [69] A. Vujic, C. Krause, G. Tso, J. Lin, B. Han, and P. Maes. 2019. Gut-Brain Computer Interfacing (GBCI) : Wearable Monitoring of Gastric Myoelectric Activity. In *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 5886–5889. <https://doi.org/10.1109/EMBC.2019.8856568>
- [70] Angela Vujic, Stephanie Tong, Rosalind Picard, and Pattie Maes. 2020. Going with our Guts: Potentials of Wearable Electrogastrography (EGG) for Affect Detection. In *Proceedings of the 2020 International Conference on Multimodal Interaction (ICMI '20)*. Association for Computing Machinery, New York, NY, USA, 260–268. <https://doi.org/10.1145/3382507.3418882>
- [71] Justin D. Weisz, Maryam Ashoori, and Zahra Ashktorab. 2018. Entanglion: A Board Game for Teaching the Principles of Quantum Computing. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '18)*. Association for Computing Machinery, New York, NY, USA, 523–534. <https://doi.org/10.1145/3242671.3242696>
- [72] Matthew Alexander Whitby, Sebastian Deterding, and Ioanna Iacovides. 2019. “One of the baddies all along”: Moments that Challenge a Player’s Perspective. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '19)*. Association for Computing Machinery, New York, NY, USA, 339–350. <https://doi.org/10.1145/3311350.3347192>